

Scope 2 Report

Summary

This proof of concept evaluates the viability of using automated online retrieval and LLM processing to fill missing parent/title metadata.

- **Objective:** Autonomously populate missing metadata fields using trusted external sources, without overwriting existing catalog data.
- **Core Finding:** The pipeline is highly viable as a *propose-then-review* tool. While strict exact-match accuracy against historical data is naturally low on multi-tag fields (e.g., matching 2 specific genres out of 5 possibilities), the model achieved a **62% High-Confidence Accuracy**, proving it successfully identifies correct, verifiable tags when tightly constrained.
- **Recommendation:** Deploy the pipeline as an assistant to the editorial team. High-confidence structured fields (like Genre) can be safely automated, while subjective fields (Theme) should trigger manual review flags.

Methodology & System Architecture

To balance token cost-efficiency with high data quality, the pipeline was engineered with strict guardrails and fallback mechanisms.

- **The Precision vs. Recall Trade-off:** The initial pipeline architecture was designed to maximize *Recall* (attempting to generate a value for every missing field). However, early iterations revealed this approach produced too much noise. Because this pipeline serves as a human-in-the-loop support tool, a false positive costs a human reviewer time to verify and correct, whereas a false negative simply leaves an empty field blank. Consequently, architecture was pivoted to optimize for **Precision**. The LLM was tuned to be highly conservative, defaulting to NULL unless the text evidence was overwhelmingly strong. This deliberate sacrifice of total coverage ensures that the proposals the system *does* generate are highly dependable, minimizing reviewer fatigue and operational friction.
- **Tiered Source Retrieval:** Rather than spamming multiple APIs, queries are routed through a hierarchical fallback chain based on source trust. Structured databases (TMDb, AniList, TVmaze) are prioritized over unstructured web scraping (Elasticsearch, Wikipedia).

- **Contextual Prompt Engineering (Cost Control):** A single, multi-field API call is made per title using Gemini 2.5 Flash. Unnecessary "thinking" tokens were capped (thinkingBudget: 256), drastically reducing costs and latency while maintaining output quality.
- **Vocabulary Forcing:** To prevent the LLM from hallucinating synonyms (e.g., suggesting "Scary" instead of "Horror"), the model is dynamically fed Shahid's exact approved vocabulary list per field and forced to output compliant JSON.
- **The "Arabizi" Transliteration Fix:** Titles written in Romanized Arabic (English letters) frequently failed standard retrieval. A dynamic transliteration loop converts these back to Arabic script, re-queries, and runs a fuzzy match-back to prevent false positives.
- **Non-Narrative Exclusion:** Sports, concerts, and ceremonies fundamentally lack narrative metadata (Plot, Era, Relation). A logic gate identifies these events and automatically flags them as out-of-scope to prevent hallucinated data.

Validation & Governance Controls

Safety and data integrity are enforced before any data is considered for publication.

- **Precision-First Gates:** The LLM must provide a verbatim text snippet ("evidence") from the source text justifying its choice. Confidence is mathematically downgraded if the token overlap between the snippet and the source is weak.
- **Trust-Weighted Ranking:** Confidence scores are tied to the source. A tag pulled from a highly trusted source (TMDb) is allowed to rank "High", whereas data scraped from Wikipedia is capped at "Medium" to mandate human review.
- **Conflict Detection:** Proposed tags are checked against any existing program-level data. If a conflict exists, the proposal is flagged and blocked from auto-publishing.

Metrics & Field-Level Recommendations

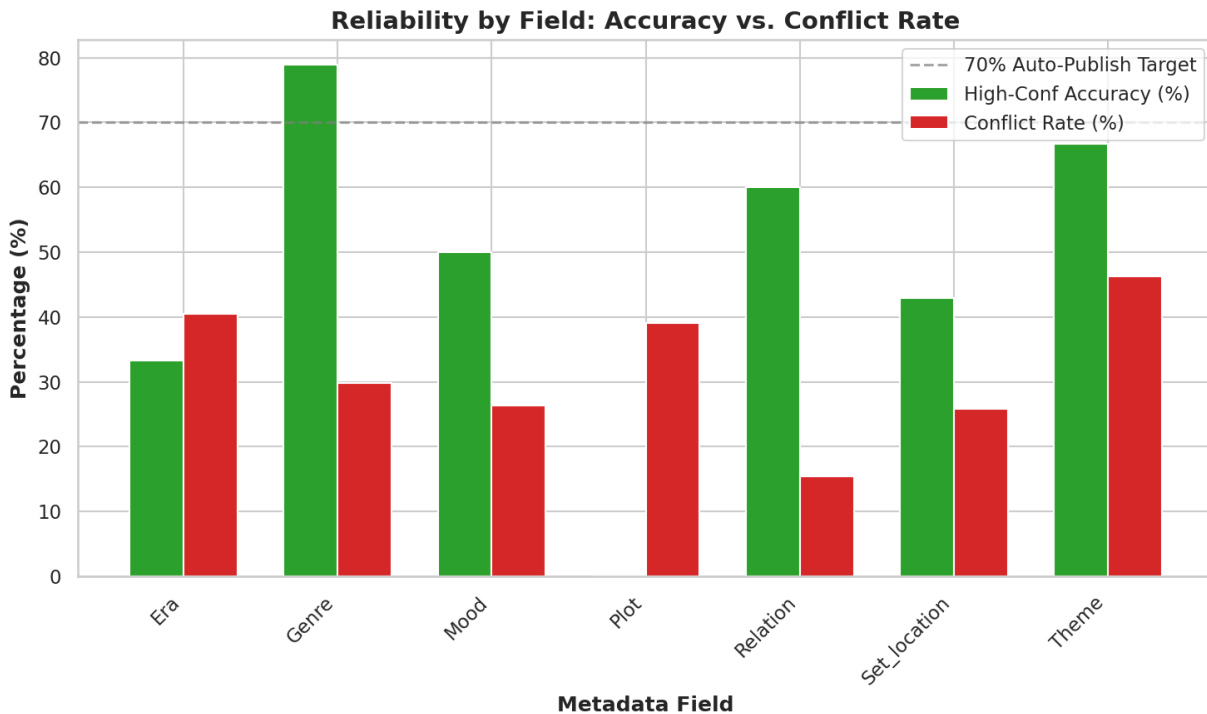
Based on a deterministic sample of test data, the fields were categorized by automation readiness:

- **Auto-Publish Candidates (High Trust):** * Genre: Reached 79% high-confidence accuracy. Safe to auto-publish when the source is structured (e.g., TMDb) and no conflicts exist.
- **Propose-Then-Review (Moderate Trust):**
 - Relation & Theme: Achieved ~60-67% accuracy. These fields are inherently subjective. The LLM accurately extracts candidate tags, but human editorial sign-off is recommended.
- **Manual-Only (Low Trust):** Dialect, Era, Plot, & Set Location: These fields suffered from high conflict rates with existing data or failed to generate enough high-confidence proposals to be deemed statistically reliable.

Visualized Metrics & Findings

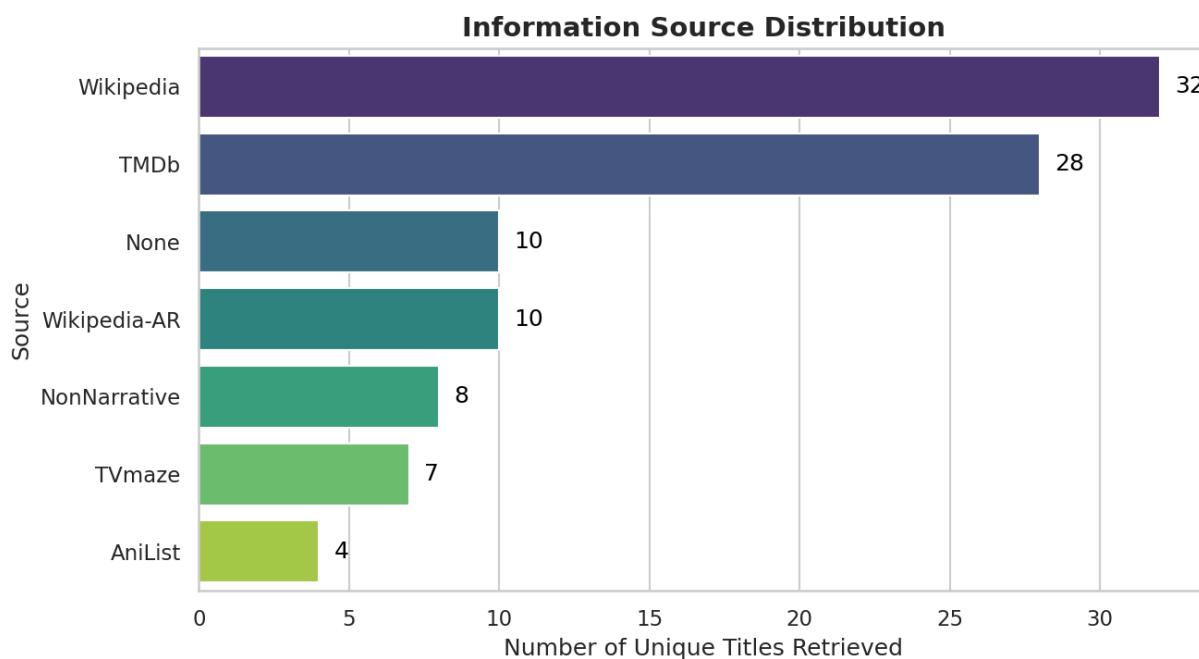
To contextualize the LLM’s performance and justify the governance rules, the pipeline generates three core visualizations based on the control and missing datasets.

Visualization 1: Reliability by Field (Accuracy vs. Conflict Rate)



- **What it shows:** A bar chart comparing the LLM's "High-Confidence Accuracy" against its "Conflict Rate" for each metadata field.
- **The Insight:** This chart proves that not all fields are created equal. It visually justifies the field-by-field recommendations. For example, it clearly illustrates why Genre (high accuracy, low conflict) is safe for auto-publishing, whereas fields like Plot or Era despite the LLM sometimes marking them as "High Confidence" carry too much conflict risk and must be routed to manual review.

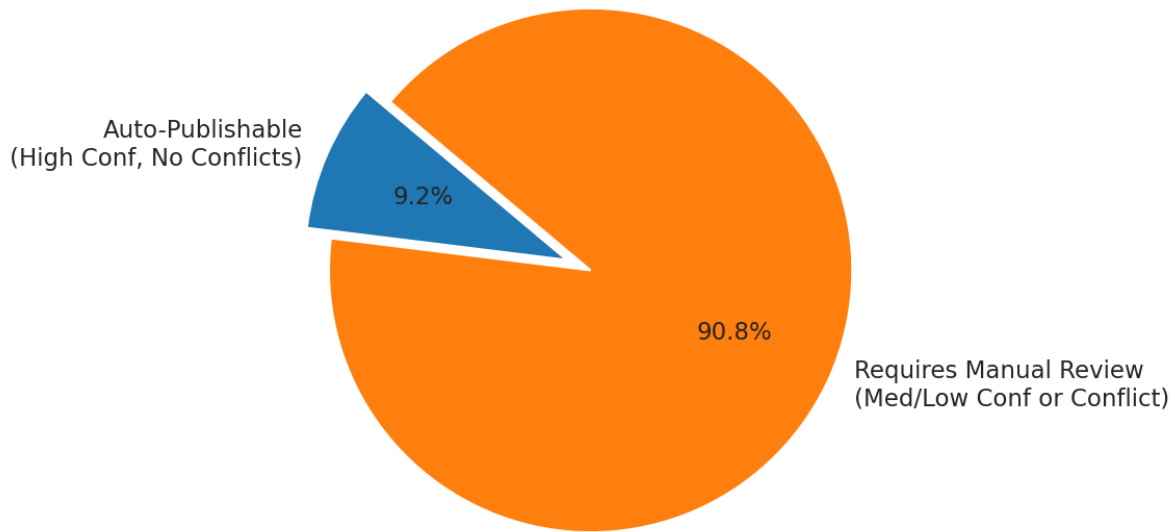
Visualization 2: Information Source Distribution



- **What it shows:** A horizontal bar breakdown of which APIs and scraped sites successfully provided the contextual text used by the LLM.
- **The Insight:** This visual validates the tiered fallback architecture. While heavy hitters like Wikipedia and TMDb naturally provided the bulk of the data, the chart shows exactly how much the specialized fallbacks (TVMaze for shows, AniList for anime, and the Arabic-transliteration loop for Wikipedia-AR) contributed. Without this multi-layered retrieval strategy, those specific titles would have remained blank.

Visualization 3: Automation Viability Funnel (Missing Data Pool)

Workflow Distribution for Missing Data Proposals (Total Fields: 131)



- **What it shows:** A pie chart isolating the *Missing Data* pool, splitting the LLM's proposals into "Auto-Publishable" (passed every strict gate: high confidence, strictly in-vocab, zero conflicts, high-trust source) versus "Requires Manual Review."
- **The Insight:** Based on the sample, roughly **9%** of missing fields can be filled completely autonomously with zero human touch. It is critical to frame the remaining ~91% not as "failures," but as **pre-filled, human-in-the-loop suggestions**. The LLM has done the heavy lifting of reading the sources and picking the tags; it simply requires a human editor to verify the subjective nuances (like Theme or Mood) before hitting approve.